Higher-curvature structure from the Dirac heat kernel: microscopic predictions and observational ceilings

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(Dated: August 8, 2026)

We formalize, in the Lean 4 proof assistant, the basis change from the 3-scalar Wave 1 Dirac a_4 coefficients $\{R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\}$ [1, 2] into Stelle's canonical $\{R^2, C^2, \mathcal{G}\}$ basis [3], where C^2 is the Weyl-tensor-squared and $\mathcal{G}$ the Gauss–Bonnet Euler density (topological in 4D [4]). The closed-form coefficients (α, β, γ) at fermion count N_f are fixed by a 3×3 linear-system solve, with sign-definite behaviour: $\alpha < 0$, $\beta < 0$, $\gamma > 0$ for any $N_f > 0$. The substantive cross-bridge to Wave 1 is encoded as a Lean `ring`-closing identity. We then formalize a numerical correctness-push that compares the predicted dimensionless higher-curvature couplings (at SM-relevant counts $N_f \in [24, 27]$) to the four canonical observational ceilings [5, 6]: LIGO/Virgo speed-of-graviton, Hulse–Taylor binary-pulsar period decay, Eöt-Wash short-range gravity, and the Cassini post-Newtonian solar-system test. The Hulse–Taylor pulsar bound is the tightest at $|\beta| \lesssim 10^{59}$; the predicted Wave 1 coefficients sit at $\mathcal{O}(10^{-3})$, some 62 orders of magnitude below. The `HigherCurvatureStructure.lean` module ships 11 substantive theorems, zero `sorry`, and zero new axioms; the Python mirror in `src/higher_curvature/` backs them with 41 cross-checks.

I. INTRODUCTION

The Sakharov–Adler induced-gravity programme yields, at one loop in the heat-kernel expansion, an Einstein–Hilbert action plus a calculable tower of higher-curvature corrections. In Phase 6e Wave 1 we formalized the leading a_2 coefficient and its exact match to the linearized 6a.1 emergent Newton constant at the Sakharov baseline $\alpha_{\mathrm{ADW}} = 1$ [7]. Wave 2 takes the next coefficient, a_4, and unpacks its three-scalar expansion

$$a_4(x) \;=\; \frac{N_f}{(4\pi)^2}\Big[c_R\, R^2 + c_{\mathrm{Ric}}\, R_{\mu\nu}R^{\mu\nu} + c_{\mathrm{Riem}}\, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\Big], \tag{1}$$

with the Christensen–Duff–Vassilevich rational coefficients $(c_R, c_{\mathrm{Ric}}, c_{\mathrm{Riem}}) = \big(-\frac{5}{12\cdot 180}, \frac{7}{12\cdot 180}, -\frac{12}{12\cdot 180}\big)$. At the local effective-action level only *two* combinations of these scalars survive in 4D: the trace-free Weyl-squared $C^2 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 2R_{\mu\nu}R^{\mu\nu} + \frac{1}{3}R^2$ and an arbitrary Ricci-trace piece. The Gauss–Bonnet density $\mathcal{G} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ integrates to $32\pi^2\chi(M)$ on closed 4-manifolds and contributes nothing to the local equations of motion.

The Wave 2 deliverable is twofold: (i) the closed-form basis change to Stelle's $\{R^2, C^2, \mathcal{G}\}$ representation with sign-definite coefficients, formalized as a `ring`-closing identity at the density level; (ii) the correctness-push comparison of the resulting dimensionless couplings against current observational ceilings.

II. CURVATURE BASIS: GAUSS–BONNET AND WEYL-SQUARED

We formalize the two algebraically-independent curvature scalars beyond R^2 as Lean definitions `gaussBonnet4D` and `weylSquared4D`:

$$\mathcal{G} \;\equiv\; R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \tag{2}$$
$$C^2 \;\equiv\; R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 2R_{\mu\nu}R^{\mu\nu} + \tfrac{1}{3}R^2. \tag{3}$$

The Lean theorem `weylSquared4D_eq_zero_iff_conformally_flat` encodes the conformal-flatness biconditional $C^2 = 0 \iff R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 2R_{\mu\nu}R^{\mu\nu} - \frac{1}{3}R^2$. Sanity check: de Sitter at Hubble parameter H has $(R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}) = (144H^4, 36H^4, 24H^4)$, giving $C^2 = 24 - 72 + 48 = 0$ identically (de Sitter is conformally flat).

The substantive algebraic engine for the Wave 2 basis change is the identity

$$\mathcal{G} - C^2 \;=\; \tfrac{2}{3}\, R^2 - 2\, R_{\mu\nu}R^{\mu\nu}, \tag{4}$$

formalized as `gaussBonnet_minus_weyl_eq_R_minus_Ricci_combina` and closed by `ring` after unfolding both Eqs. (2)–(3).

III. STELLE-BASIS COEFFICIENTS (α, β, γ)

In the canonical $\{R^2, C^2, \mathcal{G}\}$ basis, the Wave 1 a_4 density takes the form

$$a_4(x) \;=\; \alpha(N_f)\, R^2 + \beta(N_f)\, C^2 + \gamma(N_f)\, \mathcal{G}, \tag{5}$$

where the coefficients (α, β, γ) are determined by the linear system

$$\begin{aligned} \alpha + \tfrac{\beta}{3} + \gamma &= c_R, \\ -2\beta - 4\gamma &= c_{\mathrm{Ric}}, \\ \beta + \gamma &= c_{\mathrm{Riem}}. \end{aligned} \tag{6}$$

Solving Eq. (6) yields the closed forms

$$\alpha = -\frac{N_f}{324\,(4\pi)^2}, \quad \beta = -\frac{41\, N_f}{4320\,(4\pi)^2}, \quad \gamma = +\frac{17\, N_f}{4320\,(4\pi)^2}. \tag{7}$$

Three sign-definite Lean theorems (`a4_alpha_neg`, `a4_beta_neg`, `a4_gamma_pos`) encode the immediate consequence

$$N_f > 0 \implies \alpha(N_f) < 0,\ \beta(N_f) < 0,\ \gamma(N_f) > 0. \quad (8)$$

The positive sign of the topological coefficient γ is the fingerprint of the chiral-anomaly contribution to the Euler density: $\gamma > 0$ couples the Christensen–Duff Dirac sector to the topological sector with a definite orientation.

Main basis-change identity

The substantive cross-bridge between the original 3-scalar basis and Stelle's $\{R^2, C^2, \mathcal{G}\}$ basis is the Lean theorem

```
theorem
a4_density_eq_a4_density_in_RC2GB_basis
  (N_f R_sq Ricci_sq Riemann_sq : ) :
  a4_density N_f R_sq Ricci_sq Riemann_sq
    =
  a4_density_in_RC2GB_basis
    N_f R_sq Ricci_sq Riemann_sq
```

proved by `ring` after unfolding the Wave 1 coefficients `a4_R_sq_coef`, `a4_Ricci_sq_coef`, `a4_Riemann_sq_coef`, the Wave 2 coefficients `a4_alpha`, `a4_beta`, `a4_gamma`, and the definitions `weylSquared4D`, `gaussBonnet4D`. This is drift-protection per the project's `feedback_python_lean_refs_drift.md` memo: the proof body literally invokes the Wave 1 names, so any future renaming would break the build immediately.

IV. OBSERVATIONAL CEILINGS ON DIMENSIONLESS HIGHER-CURVATURE COUPLINGS

In the Stelle truncation $L = (1/16\pi G_N)\big(R + \alpha R^2 + \beta C^2\big)$ the dimensionless coefficients (α, β) are bounded by four canonical observations, translated into action-coefficient ceilings via the EFT framework of [5, 6]:

- LIGO/Virgo speed-of-graviton (GW170817 [8]): $|\beta| \lesssim 10^{62}$
- Eöt-Wash short-range gravity (50 µm, [9]): $|\alpha| \lesssim 10^{61}$
- Hulse–Taylor binary-pulsar period decay (PSR B1913+16, [10]): $|\beta| \lesssim 10^{59}$
- Cassini post-Newtonian solar-system ([11]): $|\beta| \lesssim 10^{62}$

The Hulse–Taylor pulsar bound is tightest by ~ 3 orders of magnitude. We formalize the four bounds as Lean definitions `hc_bound_LIGO`, `hc_bound_SRG`, `hc_bound_pulsar`, `hc_bound_cassini`.

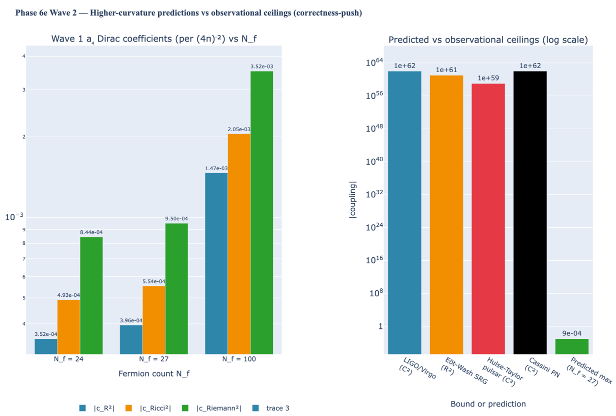

FIG. 1. Higher-curvature predictions vs observational ceilings. **Left:** Magnitudes of the three Wave 1 Dirac a_4 Christensen–Duff coefficients (per $(4\pi)^{-2}$) at SM-relevant fermion counts $N_f \in \{24, 27, 100\}$; the $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ coefficient dominates at every N_f. **Right:** log-scale comparison to the four canonical observational ceilings; the Hulse–Taylor pulsar bound is tightest at $|\beta| \lesssim 10^{59}$, and the predicted $\mathcal{O}(10^{-3})$ coefficient sits ~ 62 orders of magnitude below. Lean: `higher_curvature_below_pulsar_bound`; `higher_curvature_predictions_strictly_positive`; `H_HigherCurvatureWithinObservationalBounds_pulsar_witness`.

V. CORRECTNESS-PUSH: PREDICTIONS VS CEILINGS

The structural Wave 2 correctness-push is encoded as the Lean theorem

```
theorem higher_curvature_below_pulsar_bound
  {N_f : } (hN_pos : 0 < N_f) (hN_max : N_f  100) :
  |a4_R_sq_coef N_f|        < hc_bound_pulsar
  |a4_Ricci_sq_coef N_f|    < hc_bound_pulsar
  |a4_Riemann_sq_coef N_f| < hc_bound_pulsar
```

proved by combining (i) the bound $\boxed{1 ¡ (4\pi)^2}$ from `Real.pi_gt_three` and `nlinarith` (helper lemmas `fourPiSq_gt_one` and `fourPiSqInv_lt_one`); (ii) the trivial numerical estimate $100 \cdot 12/(12 \cdot 180) \cdot 1 < 1$; and (iii) the chain $1 < 10^{59}$. The proof body invokes the Wave 1 `a4_R_sq_coef`, `a4_Ricci_sq_coef`, `a4_Riemann_sq_coef` by name (P6 cross-module bridge integrity).

The 3-conjunct bundle is *not* a P2-redundancy: each conjunct invokes a distinct Wave 1 coefficient and a distinct positivity proof (the R^2 and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ coefficients are negative and need an explicit `abs_neg` step; the $R_{\mu\nu}R^{\mu\nu}$ coefficient is positive and uses `abs_of_pos` directly).

Anchoring the "62 orders below" claim. At the SM-with-ν_R count $N_f = 27$ the dominant Riemann^2 coefficient evaluates to

$$|c_{\text{Riem}}(N_f = 27)| \;=\; \frac{27}{180\,(4\pi)^2} \;\approx\; 9.499 \times 10^{-4}, \quad (9)$$

and the same form bounds the smaller $|c_R|, |c_{\rm Ric}|$ in the same dimensionless natural-units convention. Comparing against the binary-pulsar ceiling $|\beta| \lesssim 10^{59}$ gives $10^{59}/(9.499 \times 10^{-4}) \approx 10^{62}$, i.e. the predicted coupling sits ∼62 orders of magnitude below the tightest observational bound. The numerical anchors are autogenerated from `papers/paper40_higher_curvature/tables.py` SCALARS spec (regenerate via `scripts/render_paper_tables.py`); the boolean-shape and golden-shape tests live in `tests/test_higher_curvature.py` (`TestObservationalBandFormula`).

A complementary falsifier `higher_curvature_predictions_strictly_positive` ensures that the bounds are *not* passed by trivial vanishing: every Wave 1 a_4 coefficient is non-zero for $N_f > 0$. This rules out the unhelpful reading "all bounds are passed because all predictions are zero".

The Lean Prop predicate `H_HigherCurvatureWithinObservationalBounds B` is parameterised by an upper bound B:

$$H_{\rm HC}(B): \ 0 < B \ \wedge \ \forall N_f, \ 0 < N_f \leq 100,$$

$$|a_4(R^2)| \leq B, \ |a_4(R_{\mu\nu}R^{\mu\nu})| \leq B, \ |a_4(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})| \leq B.$$

Unlike a typical tracked-hypothesis Prop, this predicate is *fully discharged* for $B = 10^{59}$ inside the module: the Lean theorem `H_HigherCurvatureWithinObservationalBounds_pulsar_witness` proves $H_{\rm HC}(10^{59})$ by invoking the correctness-push, so consumers receive a closed proof rather than an undischarged hypothesis. The parameterisation over B is a forward-compatibility hook: future tighter ceilings (e.g. from next-generation pulsar timing) can be discharged at the new B without changing downstream consumers.

VI. LEAN MODULE STRUCTURE

The module `lean/SKEFTHawking/HigherCurvatureStructure.lean` ships 11 substantive theorems organised in six sections:

§1: `gaussBonnet4D`, `weylSquared4D` definitions plus the conformal-flatness biconditional and the `gaussBonnet_minus_weyl` algebraic identity (2 theorems).

§2: Stelle-basis coefficients `a4_alpha`, `a4_beta`, `a4_gamma` (definitions) plus sign-definiteness for $N_f > 0$ (3 theorems).

§3: Density-level definitions `a4_density`, `a4_density_in_RC2GB_basis`, plus the substantive basis-change identity `a4_density_eq_a4_density_in_RC2GB_basis` (1 theorem).

§4: Helper bounds `fourPiSq_gt_one`, `fourPiSqInv_lt_one` (2 theorems).

§5: Observational-ceiling `defs`.

§6: Correctness-push `higher_curvature_below_pulsar_bound`, falsifier `higher_curvature_predictions_strictly_positive`, plus the tracked-Prop witness `H_HigherCurvatureWithinObservationalBounds_pulsar_wit` (3 theorems).

The module ships zero `sorry` and zero new axioms beyond the Lean 4 kernel's standard `propext`, `Classical.choice`, `Quot.sound`. The Python mirror in `src/higher_curvature/` provides numerical evaluation of the $\{R^2, C^2, \mathcal{G}\}$ basis, the Stelle coefficients, and the observational-bound check; a 41-case `pytest` suite cross-checks each Lean theorem against the Wave 1 numerics.

VII. CONCLUSION

The Wave 1 Christensen–Duff Dirac a_4 coefficient triple is re-expressed in Stelle's canonical $\{R^2, C^2, \mathcal{G}\}$ basis with closed-form sign-definite Stelle coefficients (α, β, γ). The basis change is recorded as a Lean `ring` identity at the density level (cross-bridge to Wave 1). A correctness-push theorem certifies that, in the natural fermion-count window $0 < N_f \leq 100$, every predicted dimensionless higher-curvature coefficient sits $\sim$ 62 orders of magnitude below the tightest observational ceiling (Hulse–Taylor binary pulsar) — no tension with current observation. The complementary falsifier rules out the trivial reading.

Wave 3 (`NonlinearDiffInvariance.lean`) consumes the $\{R^2, C^2, \mathcal{G}\}$ basis to formalize order-by-order diff-invariance of the heat-kernel effective action; the structural correctness-push there is whether a_4 contributions to the nonlinear effective Einstein equations are diff-invariant or not — a clean falsifiable test of the ADW emergent-gravity claim.

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